

Complex Adaptive Systems

SENSS Specialist Training: Introduction to Computational Methods

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Day 2

Complex Adaptive Systems

- What do we mean by complexity in (a social system)?
- What makes a [social] system complex?

Complex Adaptive Systems

*Complexity is not just 'many parts'; it is **dependence and interactions** among parts*

Hallmarks of Complexity:

1. Interacting agents (agents respond to each other)
2. Non-linearity (and phase-transitions)
3. Heterogeneity (agents may differ in core characteristics)
4. Feedback loops (negative or positive)
5. Network topology
6. Adaptation
 - agents may change behaviour
 - influence each other / environment
7. Memory and learning

1. Interaction

Schelling's Segregation Model:

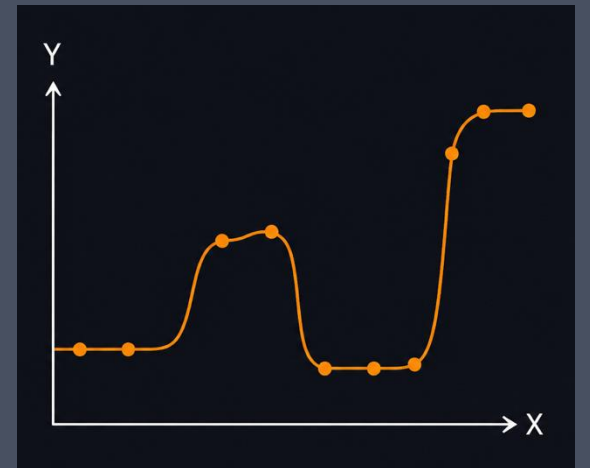
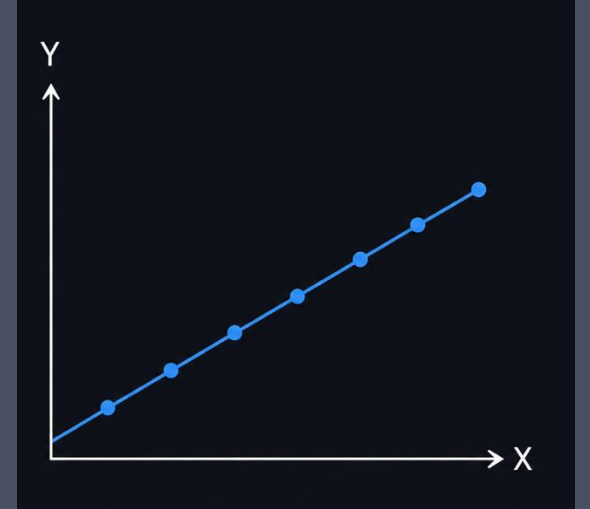
- Agents observe, influence and respond to their local neighbours
- Someone else's behaviour might affect your behaviour

Dissatisfied agents marked with *

X	X*	O	X*	O
	O	O	O	O
X	X			
X	O*	X	X	X
X	O	O		O*

2. Non-linearity

- Linear systems are orderly and consistent
 - proportional
 - small change $\rightarrow$ small effect
 - large change $\rightarrow$ large effect
- Non-linear systems: hard to guess
 - disproportionate
 - large change $\rightarrow$ no effect
 - minuscule change $\rightarrow$ large effect



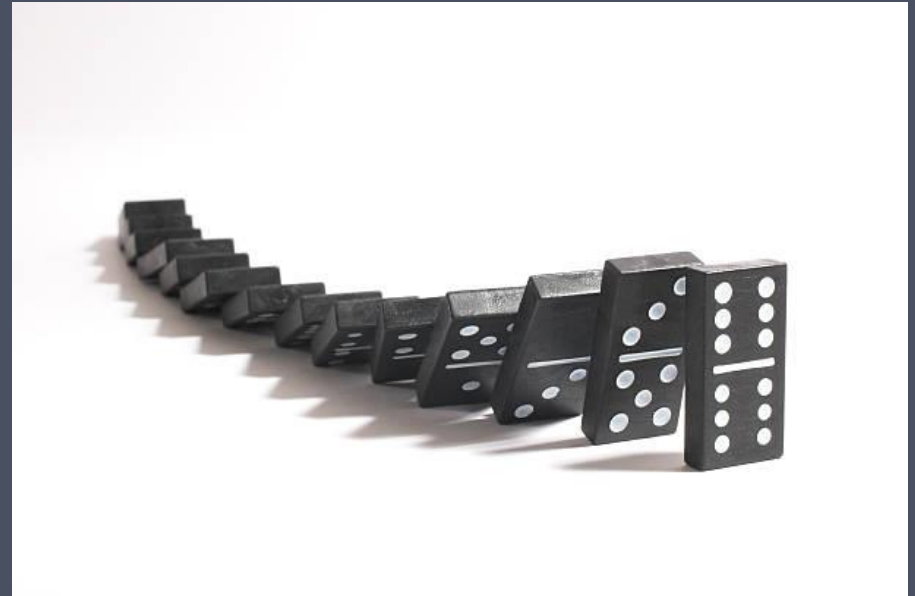
2. Non-linearity

- Tipping points or critical values
- small deviations → substantial differences
- phase transitions refer to non-linearity

Examples:

- crowd formation, rebellions, standing ovation

https://www.ted.com/talks/derek_sivers_how_to_start_a_movement

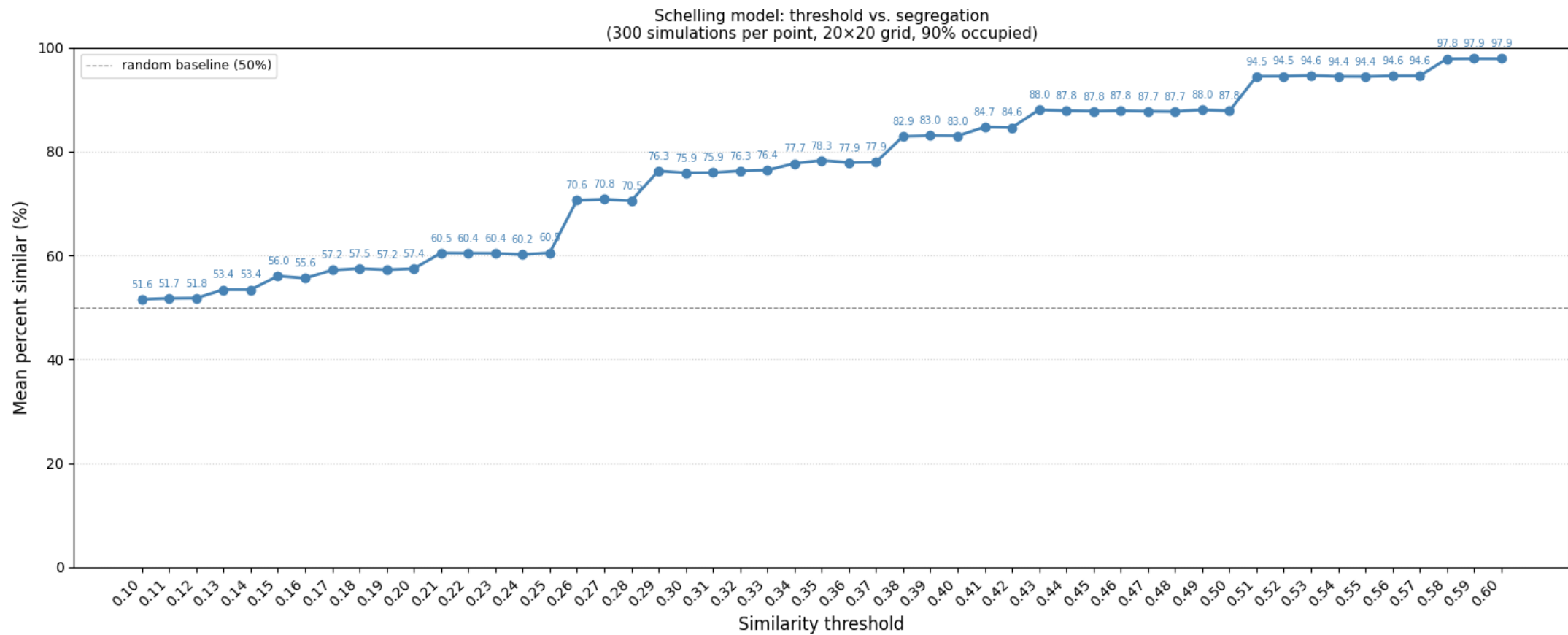


2. Non-linearity

Where is non-linearity in the Schelling's segregation model?

2. Non-linearity in Schelling's Model

- Let's run the model again and again with different thresholds
- Vary threshold from 0.1 to 0.6 by 0.01.
- Run 300 simulations for each threshold
- For each threshold, calculate the level of similarity (percent similar)
 - This is a similarity index. A metric for measuring segregation
- Take the mean of the index for the threshold
- Plot the means for each threshold



2. Non-linearity in Schelling's Model

Non-linearity:

- A *tipping point* (aka a *critical value*) in the segregation threshold τ where small changes produce large segregation
- Small changes in the threshold τ may lead to major changes in the system
- Major changes in the threshold τ may lead to virtually no change in the system
- It is a step-function: non-linearity

3. Heterogeneity

- Agents differ in attributes, preferences, and resources
 - same situation → different responses
 - Axelrod's culture model
- Environment may differ also differ
 - e.g., some grid cells are more desirable
- Heterogeneity is deliberately minimal in Schelling
 - only a single attribute (red / blue); a shared threshold

4. Feedback loops

- Negative feedback: stabilize
 - Consider the ant colony example: ants change type when too few type
 - Process counteract itself
- Positive feedback: augment changes
 - Example: bank run (or crowds/protests): exponentially join in
 - Process reinforce itself

4. Feedback loops

- Feedback → complexity
 - stability or amplification
 - tipping points and cascades (e.g., bank run)
- Differences may cancel out
 - averages are informative
- Differences may reinforce each other (complexity)
 - unexpected behaviour
 - phase transitions
- *Phase transitions*: significantly different states and dynamics

5. Network Topology

- The geography of interaction
- Connectivity
- Who is connected to whom and how
- Different network structure → Very different dynamics
 - even with the same agents & similar rules

How to Model Standing Ovation?

Simpler approach:

- Individuals stand up:
 1. if they like the show
 2. if majority of the people stand up

Network topology:

- Individuals stand up:
 1. if they like the show
 2. if they see people standing up

Vision (seat and theatre structure) now matters

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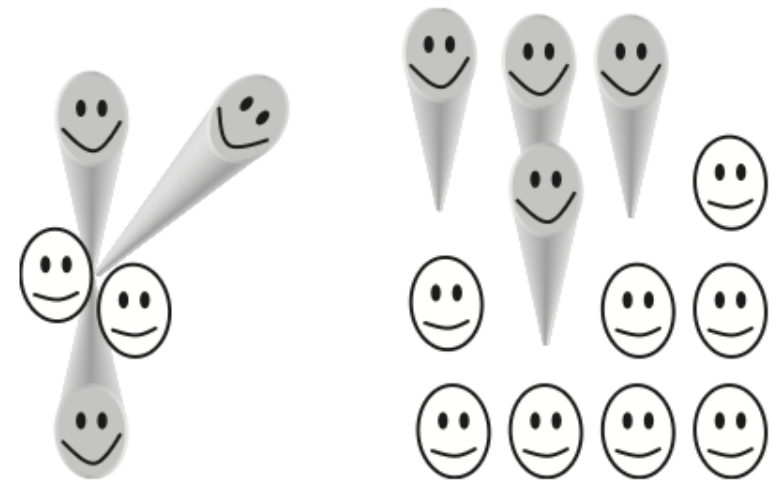


Figure 2.1. Two views of modeling the standing ovation. In its simplest form, the model requires that everyone shares the same seat in the auditorium (*left*), while the more elaborate model (*right*) allows space, friendship connections, and physical factors like vision to play a vital role in the system. While the simple model might rely on traditional tools like formal mathematics and statistics, the more elaborate model may require new techniques like computational models using agent-based objects to be fully realized.

6. Adaptation

- Agents can change in response to feedback or circumstances
 - they update behaviour as conditions change
 - they reshape the environment, which then reshapes them
- Agents are dynamic (as opposed to static)
- Adaptive $\neq$ optimal
 - bounded rationality; limited information; satisficing

7. Memory and Learning

- **Memory:** past states influence present behaviour
 - history matters!
- **Learning:** agents update their decision rules over time
 - Individual learning or learning from others
 - Learned attributes may spread across a network

Simple vs Complex Simulations

- Schelling's segregation model is sometimes called a *pet model* or a *simple social simulation*.
- Let's discuss why that is. What does it even mean? Do we agree with that assessment?

Simple vs Complex Simulations

- *Complex social simulations are characterized by numerous interacting entities, typically heterogenous in several respects, governed by multiple and typically nonlinear dynamics.*

Why is Schelling's model called simple?

Limited interactions: agents only indirectly influence each other through neighbourhood composition

Agents:

- do not communicate
- do not cooperate or compete
- cannot influence each other's preferences
 - (for example: the threshold τ)
- cannot become friends or fight with each other
- do not adapt

Why is Schelling's model called simple?

Schelling's agents are not very sophisticated:

- only a single attribute: red or blue
 - other significant attributes (e.g., wealth/income; religion/religiosity; age; how loud, etc.)
- they process only a single piece of information (neighbourhood composition)
 - agents process local information only
 - they do not process system-level information
- a single, simple rule govern their behaviour
 - they cannot strategise (e.g., wait out that the other neighbour moves first; strategically move somewhere)
- they do not have any memory; no learning

Why is Schelling's model called simple?

Schelling's agents are (largely) homogenous:

- same tolerance threshold
- same decision rule
- same type of information

How can we make them more heterogenous?

Why is Schelling's model called simple?

Limited network topology

- Agents interact only with immediate neighbours on a grid
- Social relationships beyond spatial proximity are ignored
- Possible extensions to make it more complex:
 - Agents could have social networks in addition to spatial neighbours
 - Agents may prefer to live close to friends or relatives
 - Long-distance social ties could influence decisions

Why is Schelling's model called simple?

Simple spatial structure

- All locations (grid-cells) are assumed to be equal
- No parks; sea-side; rivers; landfill/waste-disposal sites; industrial zones; schools; hospitals, etc.
- Homogeneous space
 - we could introduce heterogeneous space where location desirability (and cost) is different

Simple temporal dynamics

- The model operates on a uniform time step: it is turn-based and sequential (same pace)
- Agents act in turns at the same pace: their 'speed' of action is the same
- More complex possibilities: asynchronous agents
 - Agents acting at different speeds or frequencies
 - Agents waiting or anticipating future neighbourhood changes

Should we make Schelling's
model more complex?

Should we make the model more complex?

- It depends on the research question and your aims: the model has always a *purpose*
 - you (and your research question) define the purpose
- More complex or sophisticated does not necessarily mean better:
 - So many dimensions and interactions: extremely hard to trace properties
 - Which question will they be addressing?
- In many cases it is the opposite: simpler models are preferable and perform better
- A more sophisticated (aka complex) model can perform worse
- Simple models can be extremely useful
- Schelling's model is simple in *modelling* (in terms of structure and design), but it is extremely powerful as a heuristic tool for understanding social phenomena

Simple Modelling but Complex System

- The model itself could be called simple, but the system behaviour is complex.
- *Emergence* of segregation: macro-level segregation emerges even when individuals have only mild preferences

Highly insightful and interesting because:

- *the micro-motives and macro-behaviour fail to align*

it is far more interesting to see cooperative behavior emerge when the agents are self-interested than when the agents are presumed to be altruistic, or to see agents aggregate into cities when their goal is to be left alone.